

Supermarket Sales Analysis

Comprehensive Data Analysis Report

Project 1: Retail Domain Analysis

Report Date: September 08, 2026

Dataset: 2,000 Supermarket Transactions

Executive Summary

This report analyzes 2,000 supermarket transactions across three branches and six product categories.

KEY FINDINGS:

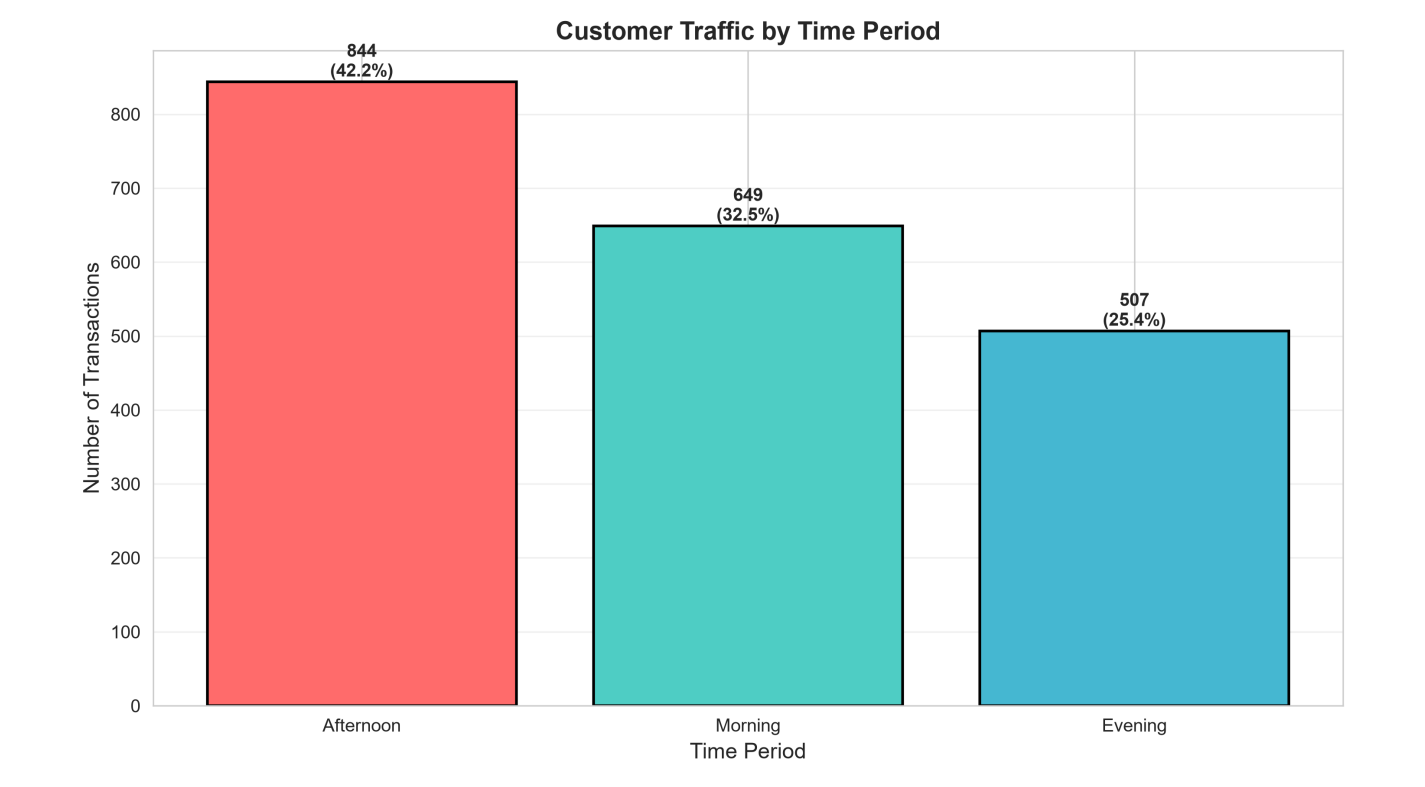
- Total Revenue: Rs. 519,280.69
- Average Transaction Value: Rs. 259.64
- Peak Period: Afternoon (42.2% of transactions)
- Best Branch: Branch B (34.8% of revenue)
- Overall Satisfaction: 6.96/10 (Moderate-to-Good)

The analysis identifies opportunities for operational optimization, particularly around staffing during high-traffic periods, branch-level performance monitoring, customer satisfaction, and seasonal sales planning.

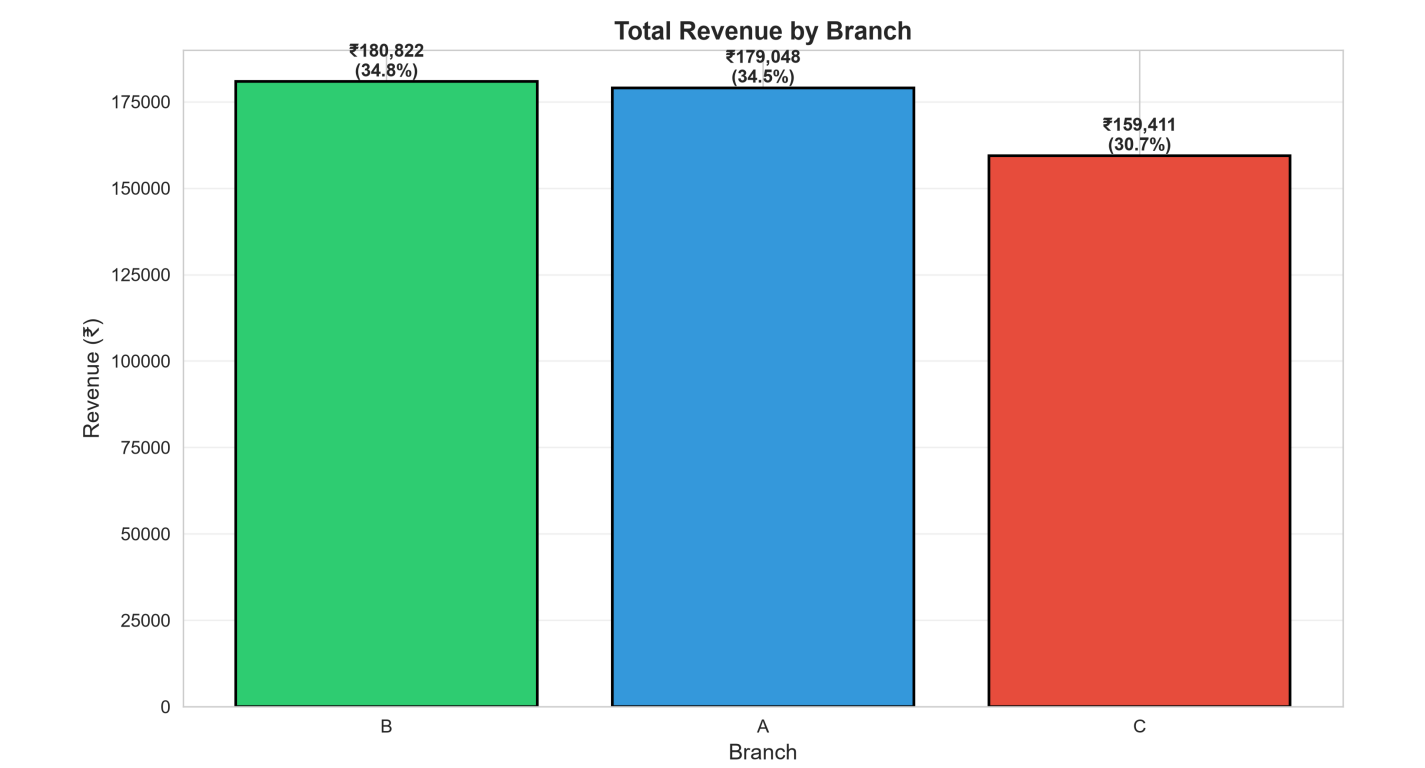
The recommendations are based on observed patterns in the supplied dataset and should be validated through operational testing and additional business data before implementation.

Key Visualizations

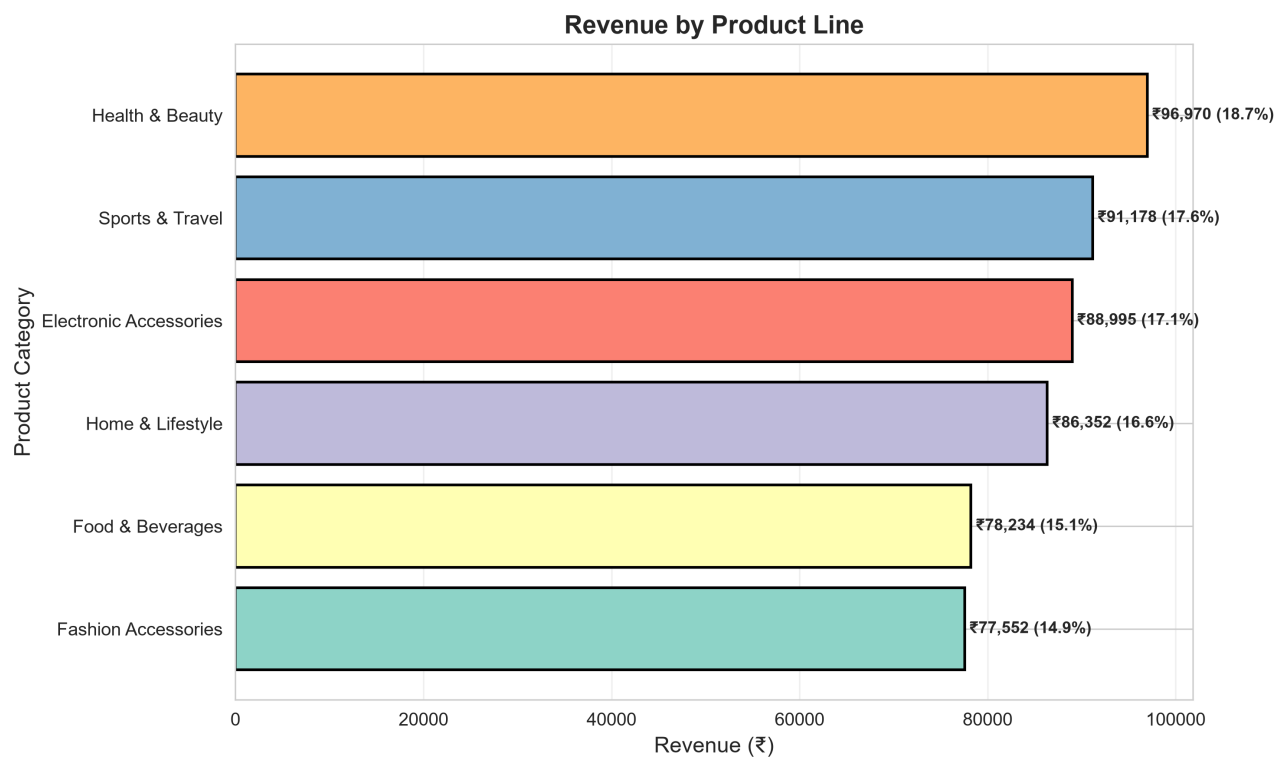
Customer Traffic by Time Period



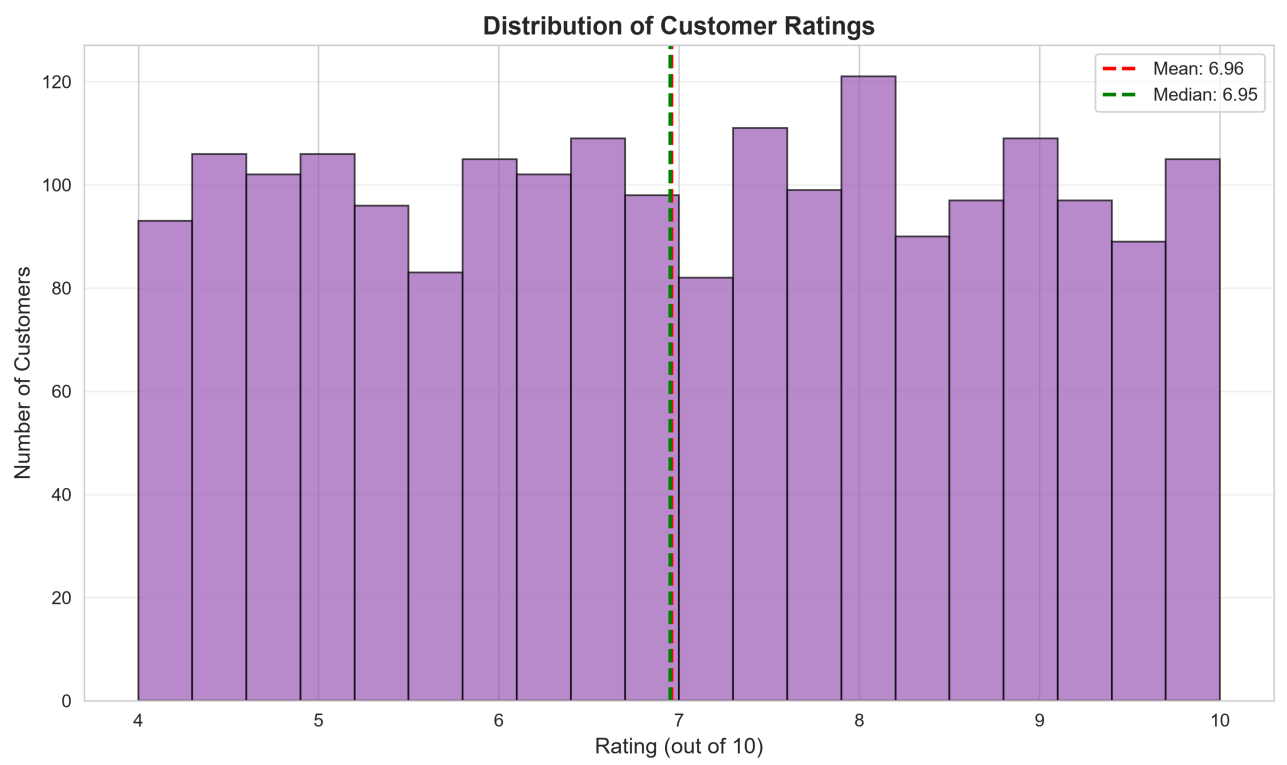
Revenue by Branch



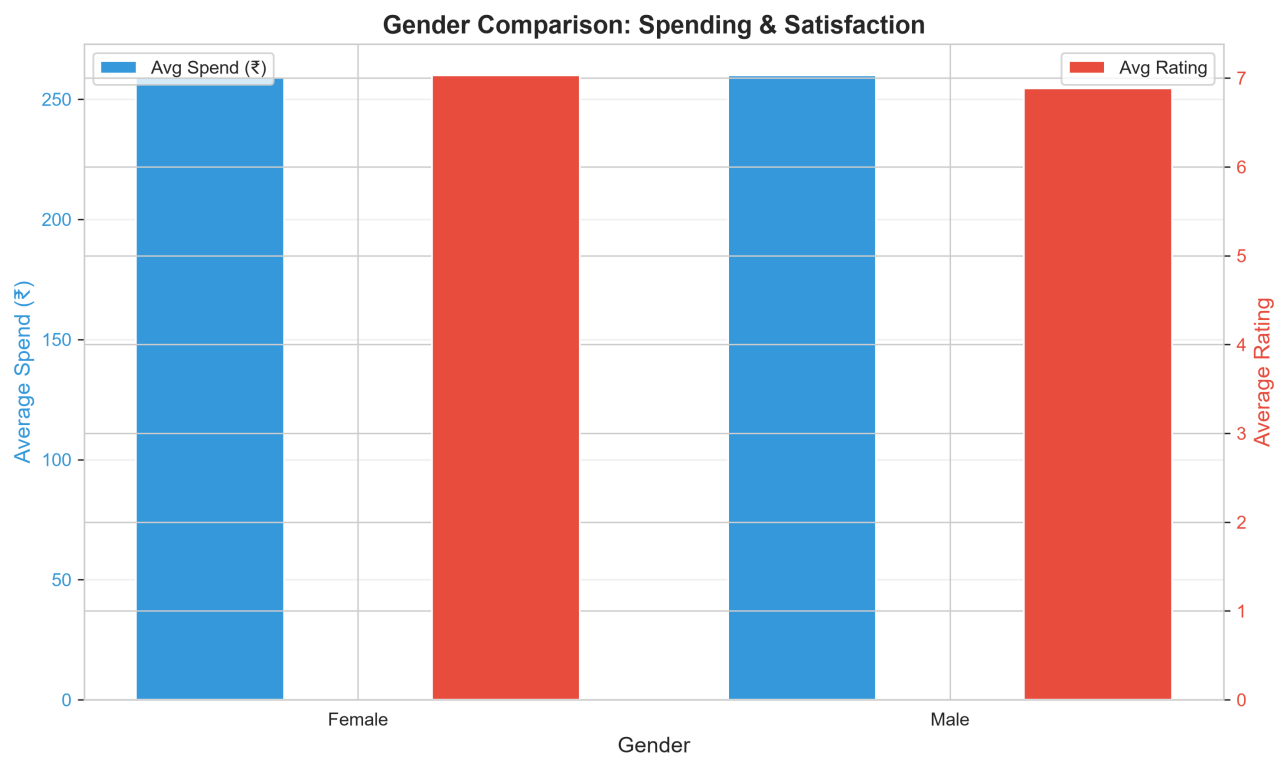
Revenue by Product Line



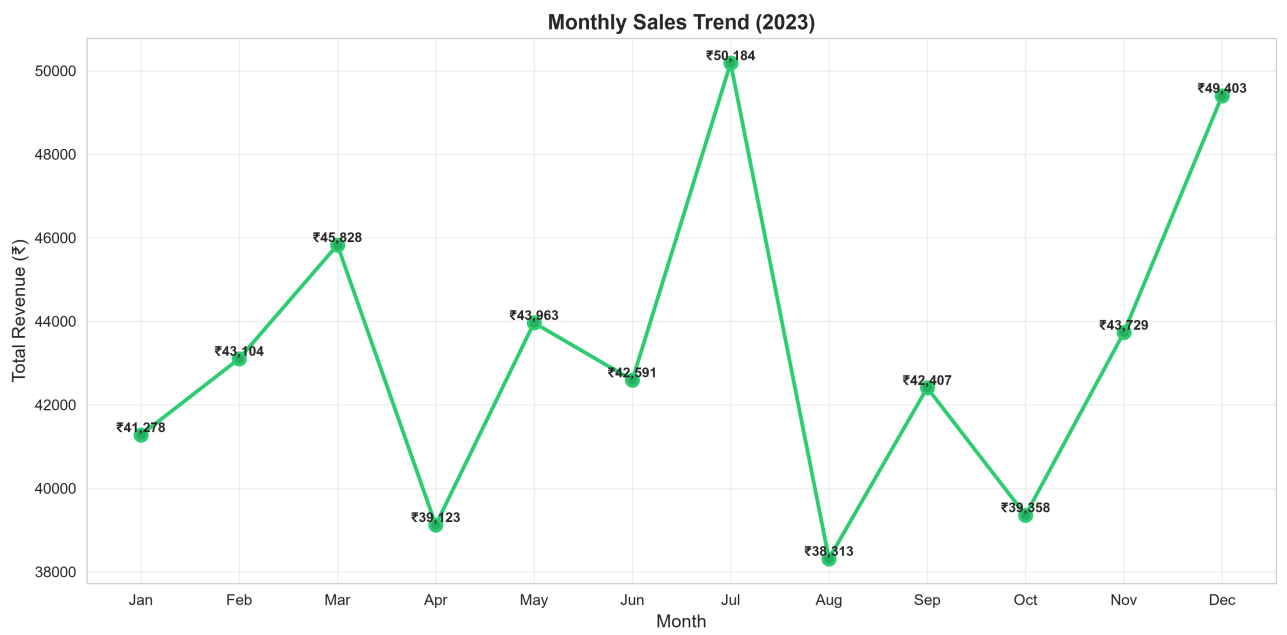
Customer Rating Distribution



Gender Comparison



Monthly Sales Trend



Key Findings

1. OPERATIONAL PATTERNS

Afternoon represents approximately 42% of transactions, indicating a high-traffic period that should be considered when planning staffing and inventory availability.

2. BRANCH PERFORMANCE

Branch B generated the highest share of revenue at 34.8%. Branch-level differences should be investigated further to understand whether they relate to customer traffic, product mix, transaction volume, or other operational factors.

3. PRODUCT PORTFOLIO

Health & Beauty generated the highest revenue share at 18.7%. The relatively distributed contribution across product categories provides a basis for category-level inventory and promotional planning.

4. CUSTOMER SATISFACTION

The average customer rating is 6.96/10. Approximately 19% of ratings fall into the low-rating segment, indicating an opportunity to investigate the factors associated with lower customer satisfaction.

5. GENDER DYNAMICS

Female customers generated approximately 3.68% more revenue than male customers in this dataset. This observed difference can inform customer-segmentation analysis, but should not by itself be interpreted as a causal effect.

6. MONTHLY SALES PATTERN

July recorded the highest monthly sales at approximately Rs. 50,184, followed by a decline in August. These observed monthly variations can be used for further investigation of seasonal or promotional effects.

Strategic Recommendations

IMMEDIATE ACTIONS:

1. Peak-period staffing

Review afternoon staffing levels using transaction volume and queue/service-time data. A controlled staffing adjustment can be tested rather than assuming a fixed percentage increase.

2. Branch performance investigation

Investigate Branch B's stronger revenue performance and compare Branch C with other branches using transaction volume, average transaction value, product mix, and customer ratings.

3. Customer satisfaction

Analyze low-rating transactions to identify recurring issues such as service experience, product availability, or checkout efficiency.

4. Inventory planning

Use the observed product-category revenue distribution to prioritize inventory monitoring for high-contribution categories.

MEDIUM-TERM ACTIONS:

5. Customer segmentation

Explore customer behavior by gender, branch, product category, and purchase patterns before designing targeted campaigns.

6. Seasonal planning

Compare monthly sales patterns with promotions, holidays, footfall, and inventory availability to determine whether the observed monthly variation is repeatable.

7. Performance monitoring

Create a recurring dashboard tracking revenue, transaction volume, average transaction value, customer ratings, and branch-level performance.

RECOMMENDED KPIs:

- Revenue by branch
- Transactions by time period
- Average transaction value
- Revenue by product category
- Low-rating percentage
- Customer rating trend
- Monthly revenue growth
- Branch-level revenue growth

The recommendations are intended as testable business actions. No specific improvement percentage is guaranteed from the available dataset alone.

Data Interpretation & Limitations

1. The dataset contains 2,000 transactions across three branches and six product categories.
2. Observed differences between branches, customer groups, or time periods describe patterns in this dataset and do not by themselves establish causation.
3. The analysis does not contain detailed operational variables such as staffing levels, queue times, inventory availability, promotions, advertising spend, or store footfall.
4. Monthly and time-period patterns should therefore be treated as observed sales patterns rather than guaranteed seasonal effects.
5. Customer-rating differences should be investigated alongside service quality, product availability, and transaction-level factors.
6. The supplied dataset contains a financial consistency issue between the provided Total value and the calculation $\text{Unit Price} \times \text{Quantity} + \text{Tax}$ for most transactions. The original Total values were retained rather than altered.
7. Additional historical data would improve the reliability of seasonal analysis and business planning.

Future analysis could incorporate staffing, inventory, promotional activity, footfall, customer demographics, and multiple years of transaction history.

Conclusion

The supermarket transaction dataset provides a useful overview of sales performance, customer behavior, product-category contribution, and time-based transaction patterns.

The analysis identifies several practical areas for further investigation, including afternoon traffic management, branch-level performance differences, customer satisfaction, product-category performance, and monthly sales variation.

Branch B recorded the highest revenue share, Health & Beauty contributed the largest product-category share, and July recorded the highest monthly sales in the analyzed period.

These findings provide a quantitative baseline for business decision-making. However, operational changes should be validated using additional business variables and controlled measurement rather than assuming fixed improvement percentages.

The recommended KPIs can be used to evaluate whether future interventions produce measurable improvements in revenue, customer experience, and operational performance.